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AWS Certified AI Practitioner (AIF-C01) Study Guide

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Tom Taulli

AWS Certified AI Practitioner Study Guide

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Tom Taulli



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AWS Certified AI Practitioner Study Guide

by Tom Taulli

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Preface

Conventions Used in This Book

The following typographical conventions are used in this book:

Italic

Indicates new terms, URLs, email addresses, filenames, and file extensions.

`Constant width`

Used for program listings, as well as within paragraphs to refer to program elements such as variable or function names, databases, data types, environment variables, statements, and keywords.

`Constant width bold`

Shows commands or other text that should be typed literally by the user.

`Constant width italic`

Shows text that should be replaced with user-supplied values or by values determined by context.

TIP

This element signifies a tip or suggestion.

NOTE

This element signifies a general note.

WARNING

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Chapter 1. Introduction

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This will be the 1st chapter of the final book.

If you have comments about how we might improve the content and/or examples in this book, or if you notice missing material within this chapter, please reach out to the editor at shunter@oreilly.com.

The AWS Certified AI Practitioner certification – which has the exam code of AIF-C01 – is focused at the entry-level for artificial intelligence (AI), machine learning (ML), and generative AI (genAI). Of course, this includes the coverage of numerous AWS tools.

According to Jenni Troutman, who is the Director of Products and Services of AWS Training and Certification:

"The AWS Certified AI Practitioner exam is a foundational AI certification that is designed for people from a variety of backgrounds and experiences to show their understanding of AI and generative AI concepts, their ability

to recognize opportunities that benefit from AI, and their knowledge on using AI tools responsibly.”

If you already have knowledge about these topics, this will certainly help with the exam.. But this is not a requirement. This book will provide what you need to know, breaking down the concepts that are covered in the exam. There are also sample questions at the end of each chapter, as well as a practice exam.

In this chapter, we’ll look at the following:

- The reasons to take the exam
- The audience for it
- Topics covered in the exam
- Other certifications to consider.

Why Should You Take This Exam?

Getting the AWS Certified AI Practitioner certification can be a boost to your career. This is particularly important in the AI industry, which is growing at a rapid pace.

Let’s take a closer look:

Recognition

The certification makes a statement. It shows employers that you understand the basics of AI, along with AWS services. You can also highlight the certification among your network, such as LinkedIn, with a digital badge for your profile.

Even if you do not plan to pursue an AI role, the certification will still be valuable. After all, when it comes to many jobs nowadays – whether in sales, marketing, legal and so on – there is usually a need to understand AI.

The Growth of AI

Gartner forecasts that spending on AI software will hit $297.9 billion by 2027, representing an annual growth rate of 19.1%. Businesses have been investing heavily in AI so as to bolster efficiency and productivity. But this powerful technology can also be a driver for growth, such as by unlocking insights from corporate data.

Of course, this means businesses will need AI savvy employees. According to a study commissioned by AWS, over 90% of employers expect to adopt AI-related solutions within the next five years. Just some of the companies that have been training employees for the AWS Certified AI Practitioner and AWS Certified Machine Learning Engineer – Associate certifications include CapGemini, Cognizant, Compass, Deloitte, DXC Technology,

EPAM, Kyndryl, N-iX, SoftwareOne, Tata Consultancy Services, and Wipro.

AWS Remains Dominant

AWS is still the dominant player in the global cloud infrastructure market. For the first quarter of 2024, its market share came to 31%. This compared to Microsoft at 25% and Google Cloud at 10%.

In the third quarter of 2024, AWS brought in $27.5 billion in revenue. This represented a 19% increase from the same period a year ago.

At the heart of the growth has been AWS's AI segment. For 2024, the revenue surged at a triple-digit pace.

Boosting Your Salary and Career with AI Skills

"Having AI skills can improve productivity," said Troutman. "In fact, 88% of workers expect to use AI in their daily work by 2028 to automate repetitive tasks and drive creativity.

In our own research, we also found that employees who have AWS certifications have improved their interactions with their technical colleagues and customers."

In light of these benefits, there has been strong gains in compensation for those with AI skills. Consider a November 2023 AWS study. It shows that their salaries are generally 43% higher for those in sales and marketing roles, 42% higher for finance roles, 41% higher for business operations roles and 47% higher for IT specialists.

Then what about those who have AI roles? Here's a look at the average compensation levels in the U.S., according to date from Coursera:

Data Analyst

This person can earn a salary from $60,000 to $90,000. This depends on a person's background and location. For example, someone living in San Francisco earns an average of $95,000.

AI Developer

This is a person – who is also called a machine learning engineer – builds, deploys and monitors AI projects and models. They understand languages like Python and various frameworks, such as TensorFlow, PyTorch, or scikit-learn. An AI developer earns anywhere from $85,000 and $120,000 per year.

Project Manager

Such a person can earn from $90,000 to $130,000 annually. Their salary depends on their skillsets and the complexity of the projects they manage.

Sales Specialist for AI Solutions

The salary is typically between $70,000 to $110,000 per year. But the compensation can be much higher because of commissions.

Note: For data scientists with extensive experience, the compensation packages can be substantial. For companies like OpenAI and xAI, they have offered salaries from $145,000 to $530,000. This does not include equity compensation or bonuses.

Improving Economic Growth and Business Operations

A McKinsey report estimates that genAI systems will add $2.6 trillion to $4.4 trillion to annual GDP for the globe. The industries that are likely to see the biggest impact include banking, high tech and life sciences.

Then there is research from Goldman Sachs, which predicts that genAI could increased global GDP by $7 trillion and productivity growth could be increased by 1.5% for the next decade. One of the firm's software analysts, Kash Rangan, noted: "Generative AI can streamline business workflows, automate routine tasks and give rise to a new generation of business applications."

In other words, businesses will need to adopt and use AI systems effectively. It could be existential for them.

Topics Covered in the Exam

The AWS exam guide provides a the topics you'll be tested on and the percentage of questions for each section. But Table 1-1 shows an overview.

Table 1-1. Topics on the AIF-C01 exam

Domain	Key Objectives
Fundamentals of AI and ML (20%)	• Define AI terms: AI, ML, deep learning, NLP, LLMs, algorithms, and bias. • Compare AI, ML, and deep learning. • Types of inferencing: batch vs. real-time. • Learning methods: supervised, unsupervised, reinforcement. • Practical applications: fraud detection, automation, scalability.
Generative AI Fundamentals (24%)	• Key concepts: transformers, embeddings, prompt engineering, diffusion models. • Use cases: chatbots, summarization, code generation. • Advantages: adaptability, simplicity, efficiency. • Limitations: hallucinations, inaccuracy, compliance challenges. • AWS tools: Bedrock, SageMaker JumpStart.

Domain	Key Objectives
Foundation Model Applications (28%)	• Model selection: cost, modality, size, customization. • Techniques for prompt engineering: chain-of-thought, few-shot, templates. • RAG (Retrieval Augmented Generation) • Customization tradeoffs: pre-training vs. fine-tuning. • Model evaluation metrics: ROUGE, BLEU, BERTScore.
Responsible AI Guidelines (14%)	• Features of responsible AI: bias, fairness, safety, inclusivity. • Tools: SageMaker Clarify, A2I, Model Monitor. • Legal risks: IP infringement, biased outputs, customer trust. • Transparency and explainability tools: Model Cards, open-source models. • Dataset diversity: inclusivity, balance, curated sources. |

Domain	Key Objectives
Security, Compliance, Governance (14%)	• AWS security tools: IAM, encryption, Macie. • Privacy considerations: threat detection, encryption at rest and in transit. • Compliance standards: ISO, SOC, algorithm accountability laws. • Data governance: lifecycle, retention, monitoring. • Best practices: secure data engineering, transparency, and training.

In this table, notice that more than half of the exam focuses on genAI (sections 2 and 3). But this should not be a surprise. This technology is often the main focus for AI projects and implementations.

Exam Details

The exam includes 65 questions, of which 50 are scored, and 15 are unscored. The unscored questions don’t impact your final score. The reason is that they are experimental. They are a way to help AWS to evaluate these questions for future exams.

Regardless, there’s no penalty for guessing. If you’re unsure about an answer, take your best shot.

The exam includes different question types:

Multiple choice

You’ll see one correct answer and three incorrect options, which are called distractors..

Multiple response

These questions have two or more correct answers out of five or more options. To get credit, you need to get each one correct.

Ordering

You’ll get three to five options that you need to arrange in the correct order for a certain task.

Matching

You’ll match a list of responses to three to seven prompts.

Case study

These questions have scenarios, which include two or more questions. Each question is scored separately.

The results of the exam are expressed as a score from 100 to 1,000. To pass, you need at least 700. The scoring system is scaled based on different versions of the exam, which have slightly different difficulty levels.

The scoring system is also compensatory. In other words, you do not have to get a passing score on each section. What matters is the overall score.

The exam lasts 90 minutes and costs $100. However, the fee may vary depending on your location.

You can choose to take the exam either online or at a test center. Either way, it is proctored. This means a person will oversee the exam to make sure there is no cheating.

To register, log in to aws.training, click on Certification in the top menu, and then select AWS Certification Account. After this, click Schedule New Exam, choose your desired exam, and select Schedule at Pearson VUE. You'll then see the scheduling page.

If you don't pass the exam, you'll need to wait 14 days before retaking it. Although, there's no limit on the number of attempts. Yet you'll have to pay the fee each time.

When you take the exam, you won't see your pass/fail status immediately. Instead, the results will show up in your AWS Certification Account under Exam History within five business days.

The certification is valid for three years.

Other Certifications

AWS has four types of certifications. First, there is the Foundational level, which does not require any prior experience. The AIF-C01 certification falls into this category.

The other ones include:

Associate

These are based on a role, such as for a developer, solutions architect or data engineer. For this type of certification, AWS recommends that you have experience in the topic.

Professional

This is for those who want to gain advanced skills for the design of applications or automated processes. AWS recommends at least two years of prior experience.

Specialty

This is when you want to dive deeply into a particular category. There are different experience recommendations that are based on the specialty.

As for what certifications to take after you have completed the AIF-C01 exam, there are several options available, as you can see in Table .

Certification	Details
AWS Certified Cloud Practitioner (CLF-C02)	• Fee: $100 • Topics covered: • Cloud concepts (24%) • Security and compliance (30%) • Cloud technology and services (34%) • Billing, pricing, and support (12%)
AWS Certified Machine Learning Engineer - Associate (MLA-C01)	• Fee: $150 • Data Preparation for ML (28%) • ML model development (26%) • Deployment and orchestration of ML workflows (22%) • ML solution monitoring, maintenance, and security (24%)
AWS Certified Machine Learning - Specialty (MLS-C01)	• Fee: $300

Certification	Details
	• Data engineering (20%) • Exploratory data analysis (24%) • Modeling (36%) • Machine learning implementation and operations (20%)
AWS Certified Data Engineer - Associate (DEA-C01)	• Fee: $150 • Data ingestion and transformation (34%) • Data store management (26%) • Data operations and support (22%) • Data security and governance (18%)
AWS Certified Data Engineer - Associate (DEA-C01)	• Fee: $150 • Data ingestion and transformation (34%) • Data store management (26%)

Certification	Details
	• Data operations and support (22%) • Data security and governance (18%)

Of course, there are many certifications that outside the AWS ecosystem:

The AI-900: Microsoft Azure AI Fundamentals

This is the foundational course for AI and Azure. It's similar to the structure of the AIF-C01 exam.

AZ-900: Microsoft Azure Fundamentals

This certification is focused on the core concepts of cloud computing – like architecture components, storage, networking and security – and Azure services. This is similar to the AWS Certified Cloud Practitioner

DP-900: Microsoft Azure Data Fundamentals

This covers Azure services for relational databases, non-relational databases and analytics workloads.

IBM AI Foundations for Business

This includes an introduction to AI and data science, as well as how to deploy AI systems in an enterprise. There are also hands-on demos using IBM Watson Studio. The online course takes about ten hours to complete and is free.

AI For Everyone by Andrew Ng (Coursera)

Andrew Ng is a leader in AI. He is the founder of Coursera, a venture capitalist and professor at Stanford University. He also cofounded Google Brain and was the chief scientist at Baidu. As for his course, it covers the fundamentals of AI, as well as how to build AI projects and even how to create a startup in the industry. It takes about six hours to complete and is free.

AWS Resources

Besides its study guide, AWS also has other value resources. For example, there are also online courses for the AIF-C01 exam. One is free and the others require a fee.

So then, why read this book if AWS already has these resources? They are certainly very useful and you should check them out.

However, with this book, you will dive deeper into the concepts. There will also be many real-world examples. Then there are the many sample

questions, as well as a glossary. The glossy is important because many of the questions are based on definitions.

Finally, for many people, they prefer to learn from a book!

Other Important Information

To take the AIF-C01 exam, you must be at least 13 years old. If you're between the ages of 13 and 17, AWS has certain requirements.

AWS also provides accommodations, such as those with disabilities. You can find details about this here.

Updates

The AIF-C01 certification will change over time. This is often due to updates in AWS services. To help you stay current, I've got a website where I'll post any updates you need to know about.

Conclusion

The AIF-C01 certification is your first step for the exciting and rewarding world of AI. Again, even if you do not want to pursue a role in this industry, the knowledge you gain from the exam will be helpful for your career.

In the next chapter, we'll get started in learning what you need to know to pass the exam.

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Chapter 2. Basics of AWS

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The launch of Amazon Web Services (AWS) – which happened in 2006 – was the result of serious problems with Amazon. The company's codebase and data systems were massive. They were also complicated to change and maintain.

Unless there was a big change, Amazon would not be able to continue to scale its growth. So CEO Jeff Bezos set forth a priority to completely rethink the company's IT (Information Technology) infrastructure.

The strategy was to break the codebase into much smaller components, called microservices. Each of these would handle a particular task, like

authentication, storage, and database updates.

With this approach, Amazon developers could work as small teams, which would allow for faster development. What emerged was the so-called “two-pizza” rule – that is, each team would have only enough members to feed with two pizzas.

But there was something else that was critical. Amazon needed to find ways to configure services, such as load balancing and managing data center operations, which could be time consuming. The new approach was to build a shared services platform. It would make it much easier to spin-up services. What this meant is that developers did not have to waste their time worrying about the complex underlying infrastructure. They could focus on coding, debugging and deployment.

All these changes became the basis of AWS and Bezos realized that many companies could benefit from this platform. He would also make it affordable for customers to use the services, by providing a pay-as-you-go model.

AWS was a big success, especially with startups that had limited services. But as time went by, larger customers would come on board.

Fast forward to today: AWS is a powerhouse, serving millions of customers. It is also a huge driver for growth and profits. A majority of Amazon’s operating income comes from AWS.

As for this chapter, we'll take a look at AWS, its services, and how to set it up. Even though the AIF-C01 exam is primarily about AI, there are still some questions about the core AWS services.

Cloud Computing

Even though cloud computing has been around for many years, the concept can still be fuzzy. Part of this is due to the pace of innovation in the industry. But another factor is the sheer number of capabilities, functions and use cases for cloud computing. For example, there are the enterprise services from platforms like AWS, Microsoft Azure and Google. Then there are the many applications, such as iCloud, Google Drive, and Microsoft's OneDrvie.

Yes, it seems that the cloud is everywhere.

But for the purposes of the AIF-C01 exam, it's important to get a good definition of cloud computing. So let's see how AWS defines it:

> *"Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider like Amazon Web Services (AWS)."*

Let’s break this down. First, cloud computing is on-demand, which means that you can access computing services whenever you want. This is done either through the AWS portal or CLI (Command Line Interface).

These computing services are virtualized. This is where physical hardware, like a server, is shared by multiple users. This is done by using sophisticated technologies like hypervisors and containerization. These essentially created isolated environments, with their own operating systems (OS’s). This makes it seem like you have your own dedicated system, which you can configure for your requirements.

Here are other advantages of cloud computing:

Lower costs

As a rule-of-thumb, the pricing is based on the usage of the computing services. There is no need to take on the upfront costs for a data center, with expensive servers, networking systems and storage.

Scalability

AWS has seemingly unlimited resources. For example, if your website has a surge in traffic, you can easily spin-up new resources to handle it.

Global footprint

AWS has data centers across the globe. Because of this, you can access your cloud computing resources near your location. This helps to reduce costs and latency (this is the time it takes for data to go from one point to another).

High availability

This refers to a system's ability to work with minimal downtime. AWS provides high availability through redundant infrastructure, monitoring, backup/recovery systems and scalability.

Security

A cloud computing system not only leverages modern cybersecurity approaches – like firewalls and data encryption – but also ongoing updates. You will also benefit from top security experts who will help to protect the platform.

Innovation

With cloud computing handling so many complex and expensive tasks, your organization will have the time to focus on more important matters, such as building better applications.

Cloud Models

There are three models for cloud computing:

- Public cloud
- Private cloud,
- Hybrid cloud

None are necessarily better than the other. Rather, each serves different types of requirements.

Public Cloud

The public cloud is the most common. It is what's at the core of AWS, as well as Microsoft's Azure and the Google Cloud Service. The cloud provider builds and maintains the infrastructure and compute resources. These are then shared among the customers, which is called the multi-tenant model. This allows for more efficiency and lower costs.

However, the multi-tenant model does have drawbacks. You are relying on a third-party for critical operations. There is also the concern about security. Then again, AWS does have strong systems in place for this. They also have 24/7 monitoring of its infrastructure.

Private Cloud

The private cloud is also known as on-premises. This uses the single-tenant model. This means that an organization builds and manages its own IT infrastructure. Depending on the arrangement, this may mean that it owns

its own data centers or leases them from a third party. Regardless, the organization remains in complete control of the operations.

This has several advantages. One is that the private cloud can be customized. An example is that an organization can use it for specialized operations, say for an automated warehouse where there needs to be low latency.

A private cloud is often the option for regulated industries, such as in healthcare and financial services. They need high levels of security, compliance and privacy. A security breach could result in expensive lawsuits and damage to a company’s brand. In light of this, the private cloud is often the best approach.

AWS provides various services for private clouds. An example is the Amazon Virtual Private Cloud (VPC), which allows for launching AWS resources in an on-premises environment.

Then what about the drawbacks of the private cloud? The main one is the cost. Of course, you’ll need to buy and maintain hardware, which can quickly become obsolete. You’ll also need a staff of IT professionals to manage the infrastructure.

Hybrid Cloud

The hybrid cloud blends the public cloud with the private cloud. This provides for sharing of data and applications among the two platforms. This may also include multicloud environments. This is where an organization uses more than one public cloud system, which can allow for more redundancy. Yet there is still an on-premises environment.

Sounds complex? It certainly is. These implementations require sophisticated IT expertise.

But AWS provides numerous tools to help out. AWS Outposts allows you to operate AWS infrastructure for on-premises environments that allow for seamless experiences. For example, a company can develop and test applications on the cloud system – which is more affordable – and then deploy them in the private cloud.

Keep in mind that the growth of the hybrid cloud market has been strong. According to research from SkyQuest Technology, the category is supposed to go from $97.8 billion in 2023 to $348.14 billion by 2031.

For the three cloud models, Table 2-1 shows the comparisons.

Table 2-1. Cloud Models

Feature	Public Cloud	Private Cloud	Hybrid Cloud
Definition	• Shared multi-tenant model	• Single-tenant model	• Combination of the public cloud and private cloud
Management	• No infrastructure management required by the user •	• Managed internally by the organization • Requires IT expertise •	• Managed by internal IT teams with external cloud provider support
Cost	• Pay-as-you-go pricing • Lower upfront costs • No hardware maintenance	• High upfront costs • Requires investment in hardware	• Can realize lower costs by balancing workloads across public

Feature	Public Cloud	Private Cloud	Hybrid Cloud
	•	and maintenance •	clouds and private clouds

Cloud Service Types

The phrase “as a service” is certainly pervasive. It applies to seemingly any type of IT service. Just some examples include Storage as a Service (STaaS), Monitoring as a Service (MaaS) and Artificial Intelligence as a Service (AIaaS).

If anything, the concept has become a key part of the marketing playbooks of many software companies. It’s also susceptible to considerable hype.

Regardless, the concept of “as a service” is fairly straightforward. It’s about renting or leasing an IT service – not owning it. That is, you generally agree to a subscription. You also can usually cancel at any time, for any reason. This flexibility has been a big driver for the popularity of “as a service” solutions.

But the market is highly competitive. This is why there are often sweeteners and promotions. For example, AWS may offer a big discount if you are

willing to sign up for a commitment of a year or more.

As for the cloud, there are three types of “as a service” flavors, which are also known as cloud service types:

- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)

Let’s take a more detailed look at the cloud service types and what AWS provides.

Infrastructure as a service (IaaS)

IaaS is about the core IT resources of a cloud platform. These include virtual machines (VMs), storage and networking. They are all virtualized services.

A VM is at the heart of IaaS. It’s essentially a software-based version of a physical computer or server. It’s as if you have access to a dedicated machine. It has its own OS, applications and hardware, such as CPU, memory, storage and networking.

In AWS, the VM is called an Elastic Compute Cloud (EC2). To see how it works, let’s take an example. Suppose you are leading a team that is developing a new ecommerce application and there needs to be a testing

environment. With AWS, you can quickly spin up the necessary infrastructure for this by creating an EC2 instance. Setting this up is a matter of filling out a form, which you can see in Figure 2-1.

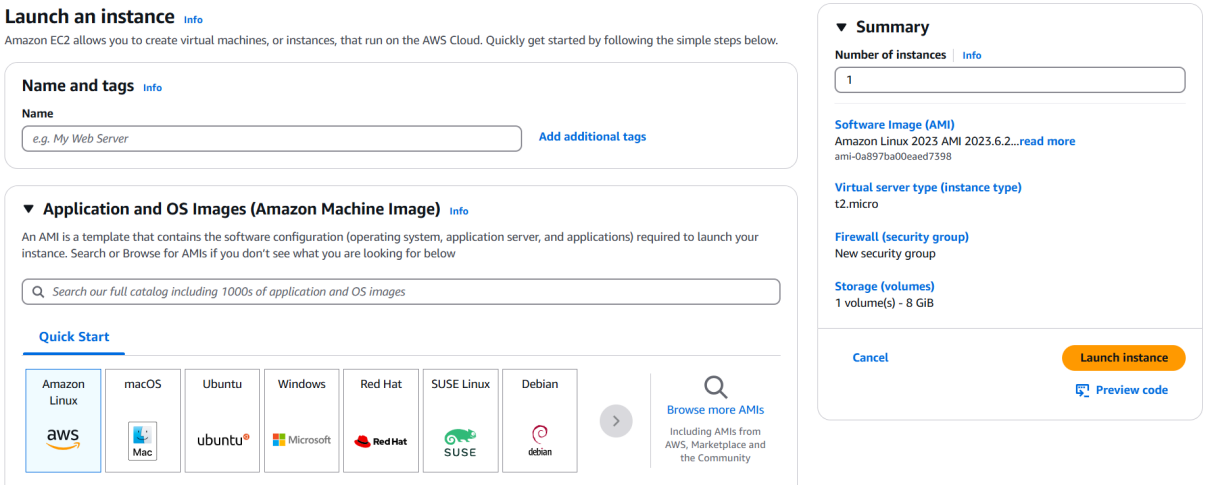


Figure 2-1. This is the configuration form for an EC2 instance.

For this form, there are several different sections. You will find them by scrolling down the page.

Here's a look:

Name and Tags

You should create a descriptive name for the EC2 instance, such as by specifying the environment, department or project.

Application and OS Images (Amazon Machine Image)

An AMI has the necessary information to launch an EC2 instance, such as with the OS and any required software. An option for this is

Quick start. But you can also use one that you have already created or that has been shared with you. You can even access an AMI from the AWS marketplace, as well as the AWS community.

Instance Type

This is information about the hardware configuration for the EC2 instance. The parameters include the CPU, memory, storage and network capacity. There are many instance types available, which you can select from a drop-down menu. They offer options for different goals, like memory or storage optimization.

Key Pair (Login)

This allows for secure logins to the EC2 instance. This is done by using key pairs of a public key (this is stored by AWS) and a private key (this is what you store). You can use an existing key pair or create a new one.

Network Settings

You can create a VPC (Virtual Private Cloud), which is a virtual network that contains the EC2 instance. You can also configure a subnet for it that has a range of IP addresses. You can then auto-assign public IPs for public addresses and create a virtual firewall, which controls inbound and outbound traffic.

Configure Storage

You can set the storage volumes for the EC2 instance, such as the root volume (this holds the OS) and optional volumes. You can set the size, type (SSD and hard drive), and encryption.

Advanced Details

This provides for various options. Some include assigning roles and permissions, automation tasks, monitoring and shutdown behavior.

After you finish the configuration for the EC2 instance, AWS will create it within a few minutes. By comparison, if you did this on your own, it could have easily taken hours – assuming you have skills with IT infrastructure – and cost thousands of dollars.

But IaaS has drawbacks:

IT expertise

You need a skilled team to manage tasks like OS updates, security patches and resource optimization. They will also need to do system monitoring and handle security.

Cost management

The pay-as-you-go model can sometimes result in unexpected expenses, such as due to over provisioning of services. When using IaaS, there needs to be strong monitoring of costs.

Vendor lock-in

Migrating from one IaaS provider to another can be expensive and time-consuming. This is why there should be strong vetting when evaluating a provider.

Platform as a Service (PaaS)

PaaS (Platform as a Service) includes the infrastructure of IaaS. But there is also middleware, which is the software that operates between the OS and the applications. These are the application servers, database management systems, messaging applications and authentication. PaaS will allow for the communication between these services.

Besides middleware, there are other services provided. You get access to development tools, like IDEs (Integrated Development Environments), as well as data encryption, load balancing, automated updates, monitoring and analytics.

Some of the key advantages for PaaS include the following:

Focus

Without the need for managing infrastructure and backend resources, developers can devote more time to building applications.

Costs

The expenses tend to be lower than IaaS, since there is no need to pay for licenses for numerous software applications. You also do not have to have staff for infrastructure management and updates.

Full-Fludged Development Platform

PaaS provides tools to help streamline the lifecycle, including building, testing, managing and updating applications. This is all done in the same integrated environment.

Analytics

This helps to test and monitor applications. This can provide for forecasting, evaluating ROI (return on investment), and improving the product design. Analytics tools also can benefit from sophisticated AI technologies.

Data Integration

PaaS can allow for real-time data operations among applications, say with ERP (Enterprise Resource Planning) and CRM (Customer Relationship Management) platforms. This allows teams to get accurate and update-to-date information to make better decisions.

To better see how PaaS compares to IaaS, let’s see an example of a PaaS service in AWS, which is Elastic Beanstalk. Suppose you want to create an application for note taking. For this, you will use Node.js and Socket.IO. By using Elastic Beanstalk, you will put the coding in app.js, the dependencies

in package.json and create a folder for the configuration. After you upload this, Elastic Beanstalk handles everything like creating an EC2 instance, load balancing, autoscaling, and SSL management.

If this were done using IaaS, the process would have been more detailed. You would need to configure the EC2 instance, setup the VPCs for the correct subnets, install Node.js and Socket.IO, create the security groups, put together the load balancing and so on. Then there will be the need for maintenance of these systems, such as with installing security patches.

This is not to imply that PaaS is better than IaaS. Rather, each approach is based on your requirements. If you want much more control over your resources, then IaaS is the best option. But if you do not have the resources or need for this level of control, then PaaS should be fine.

Software as a Service (SaaS)

SaaS (Software as a Service) means that the cloud provider manages the services for IaaS and PaaS. The user only needs to worry about using the application. For example, when you are using Gmail, Google Docs or Slack, you are using a SaaS app.

Usually, these are accessed from a web browser. But it is becoming more common for there to be versions for the desktop.

In many cases, the pricing model is based on subscriptions per user. This could be on a monthly or annual basis (for this, there is often a discount for making a one or two year commitment).

AWS provides a variety of SaaS applications for helping with business operations:

Amazon WorkDocs

This provides for secure document sharing and collaboration. This has encryption for content that is at transit or at rest.

Amazon Chime

This is a communications system, allowing for video conferencing, chat, and screen sharing.

Amazon Connect

This is a fully-managed contact center. In just a few clicks, you can setup the system.

Amazon Q

This is an AI-powered assistant. It can not only answer questions, but generate content like emails and automated workflows.

Amazon Q Developer

This leverages generative AI for software development. It can generate, debug, test and deploy software code. Amazon Q Developer also has security scans.

Regarding the three cloud types, Table 2-2 shows how they differ.

Table 2-2. The Cloud Types

Feature	Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)	Software as a Service (SaaS)
Services and Resources	• Core IT resources like VMs, storage, networking • Example: AWS EC2	• Includes IaaS infrastructure plus middleware • Provides development and deployment environment • Example: AWS Elastic Beanstalk	• Ready-to-use applications • Examples: Amazon WorkDocs, Amazon Q
Management	• User manages OS, middleware, applications	• Provider manages infrastructure and middleware	• Provider manages everything

Feature	Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)	Software as a Service (SaaS)
	• Requires IT expertise for updates and monitoring	• User manages applications • Reduced management overhead	
Customization	• High flexibility and control • Full infrastructure configuration • Custom security and networking options	• Moderate flexibility • Limited customization	• Configuration within app

AWS Regions, Availability Zones and Local Zones

AWS's global cloud infrastructure is organized into three main categories:

- AWS regions
- Availability Zones
- Local Zones

They allow for bolstering high availability, disaster recovery, low latency, security and lower costs.

Let's take a look at each.

Region

A region represents a specific geographic area. Currently, there are 36 regions across the globe, which you can see in Figure 2-2.



Figure 2-2. These are the AWS regions across the globe.

A region has a name that is based on the location. For example, “us-east-1” is for the main US East region and “eu-central-1” is for central Europe.

Each region is isolated from the others, which allows for much more stability, such as if there is a power outage or malfunction.

When selecting a region, these are some factors to think about:

Proximity to users

Having a closer region can help to speed up the delivery of the content. It can also help to lower costs.

Compliance

In certain countries, you may have data residency requirements. This means the data cannot leave the geographical bounds. For example, suppose this is the case for a healthcare company based in Australia. They will need to use the “ap-southeast-2” region to be in compliance.

Available services

A closer region may not have the AWS services you need.

Costs

The pricing can vary from one region to another. This may also include the impact of taxes in certain countries.

Availability Zones

For AWS, each region has at least three availability zones (AZ’s), which include multiple data centers. Currently, there are 114.

An AZ has its own cooling power and physical security. Then there are connections using ultra-low-latency networks. To further improve latency, the AZs are located tens of miles from each other. The is generally enough to mitigate the likelihood of several of them being impacted from an outage or disaster. However, if you have the need for high levels of security, you can run applications across more than one AZ.

Local Zones

A local zone is within a region. It allows for services to be closer to users. This is for applications that need extremely low latency, such as for streaming media, gaming, Virtual Reality (VR) and AI.

A local zone can also be effective for data residency, as well with running hybrid cloud environments.

Pricing Models

AWS provides numerous pricing models:

On-Demand Instances

As we mentioned earlier, this is about the pay-as-you-go approach. For Linux instances, the billing is on a per-second basis, whereas Windows is done hourly.

Savings Plans

This gives you discounts when you commit to a certain amount of usage – in terms of dollar-per–hour – for one to three years. There are two options available. You can have a savings plan for EC2 instances, which are limited to a region. Next, there are spot instances. This is when you access unused AWS capacity. In some

cases, the savings can be up to 90% when compared to on-demand pricing. But there's a hitch: AWS can take back resources within a two-minute window. This is why spot instances are mostly for tasks that are not mission critical.

Dedicated Hosts

This option is when you are accessing dedicated physical servers. Often this is when you have stringent compliance requirements. You can also use existing software licenses, which can help to lower the costs.

When it comes to pricing, there are many factors at play. To help, you can use the AWS Pricing Calculator. You will input details about services you plan to use, the regions they will operate in, and the expected usage.

You will then get a report. However, it is an estimate. The costs can easily change based on how you use the services.

Core AWS Services

AWS gives you access to more than 200 services. All of them are packed with many helpful features.

Some of the core services include:

- EC2

- Amazon Elastic Block Store (EBS)
- Simple Storage Service (S3)
- Lambda
- Amazon RDS
- Amazon DynamoDB
- Amazon CloudFront
- AWS Glue
- Amazon Redshift
- AWS CloudWatch

Let’s take a look at each.

Elastic Compute Cloud (EC2)

While we already discussed EC2 instances, it’s important to add some more details. This is an incredibly important topic for AWS – and the exam.

First of all, a key advantage of an EC2 instance is its scalability. There are two options for this:

Vertical scaling or scaling up/down

To handle higher workloads, you will need to add or reduce resources – like CPU power and memory – to an existing EC2 instance. The drawback is that you will usually need to stop the instance and restart it. This can mean some downtime.

Horizontal scaling or scaling out/in

This involves increasing or decreasing the number of EC2 instances. AWS can do this automatically, based on changing demand patterns. You can also set minimum or maximum EC2 instance counts.

As for accessing an EC2 instance, there are several approaches:

AWS Management Console

This is a web-based interface (we'll describe this later in the chapter). You will choose the instance and click "Connect" to access it.

SSH

You use this to securely access Linux-based EC2 instances. SSH is a protocol that requires setting up a key pair.

EC2 Instance Connect (EIC)

You access a Linux-based EC2 instance from a browser-based terminal. It's easier than having to manage SSH keys.

RDP (Remote Desktop Protocol)

This is a graphical interface to remotely access a Windows-based EC2 instance.

AWS Systems Manager

> This provides for secure connections to an EC2 instance without having to open inbound ports or using SSH keys. It’s particularly useful when you are not working with public IP addresses.

Managing EC2 instances can be tricky. Costs can easily spiral out-of-control if there is not an effective setup and monitoring.

To help out, AWS has the Compute Optimizer. This allows you to optimize your EC2 resources for your expected workloads. A key feature is rightsizing. This makes it possible to customize preferences, such as for CPU and memory usage thresholds as well as instance types.

The Computer Optimizer also comes with integration with monitoring tools. Some examples include Datadog, Dynatrace, Instana, and New Relic.

Amazon Elastic Block Store (EBS)

Amazon Elastic Block Store (EBS) is a high-performance storage service for EC2. A key benefit is that it is persistent. This means that the data will be maintained even if the EC2 instance is stopped or terminated.

EBS is also fast and scalable. Then there is the high availability, which provides for replications within AZs and 99.999% durability.

There are different versions of EBS. For when speed is the critical factor, there are SSD-backed volumes like gp3, gp2, io1 and io2. Or, when cost is

more important, then there are HDD-backed volumes, such as st1 and sc1.

Some of the use cases for EBS include:

- Disaster recovery
- Machine learning and AI projects
- Complex storage tasks on platforms from SAP, Oracle and Microsoft

Simple Storage Service (S3)

Amazon Simple Storage Service (S3) storage organizes your data into buckets. This not only includes files but also metadata, which is the description for data. This includes when the file was created, the size and settings. AWS will handle some of this automatically.

Here are some of the benefits of S3:

Durability

S3 is designed to protect your files with 99.999999999% durability. The main reason is that the data is stored across different devices and locations within an AWS region.

Scalability

You can scale up or down as your needs change. In fact, there is seemingly unlimited storage capacity. You will never run out.

Security

You can control access to S3 instances. You can do this with bucket policies and access control lists for specific files. The data is also encrypted.

S3 is incredibly versatile. Of course, a common use case is with backups and disaster recovery. But S3 is also useful for hosting simple websites that have HTML, CSS and JavaScript. You can even use S3 storage for sophisticated data analytics.

S3 gives you various storage classes. They can help you balance cost and performance:

S3 Standard

This provides quick access to files, with low latency. This is when you use them often.

S3 Intelligent-Tiering

AWS automatically transfers your data between high and lower cost storage. This is based on the usage frequency.

S3 Standard-Infrequent Access (Standard-IA)

This is a good option for the data you do not use often but want ready access to.

S3 One Zone-Infrequent Access (One Zone-IA)

This is data storage in one location. This is a low-cost option.

S3 Glacier Classes

These are generally for archived files. You can have this for a range of options, from instant retrieval or for when you rarely need access to it.

Lambda

AWS Lambda lets you write code in many languages like Python, Node.js., Java and Go. You create functions that are activated when certain conditions are triggered. It also scales automatically.

When using the tool, you are only charged for the usage of your functions and how long they take to run. But AWS has a generous free tier. This is for one million requests per month.

Lambda is an example of serverless. This means that a service will handle many of the requirements for the underlying system, such as setting up the servers, security, load balancing and on.

Amazon RDS

Amazon RDS (Relational Database Service) is a managed relational database. It is built to make it easier to setup, use and scale a system.

A relational database stores data into tables, which are made up of rows and columns. Each of the rows is a record and the column represents a feature of the data. The tables are linked by using primary and foreign keys, which allows for sophisticated analysis, data retrieval and data integrity. There is also a scripting language for this, which is called SQL (Structured Query Language).

Amazon RDS has a version of a relational database from AWS. It’s called Amazon Aurora and is compatible with MySQL and PostgreSQL.

But you can use other popular third-party databases too:

- Oracle
- Microsoft SQL Server
- MySQL
- MariaDB
- PostgreSQL

For all of these, you get the following benefits:

- Automatic backups.

- Security features like VPCs and encryption.
- Monitoring that is integrated with Amazon CloudWatch.
- Deployments across multiple AZs.

Amazon DynamoDB

Amazon DynamoDB is a serverless NoSQL database. Unlike a relational database, a NoSQL database is designed for unstructured or semi-structured data. This is where the information is not organized in rows and columns. A NoSQL database will store information based on approaches like key-value collections, document format (like in JSON or XML) or relationships (called graph databases).

Some of the use cases are:

- Real-time trading
- Fraud detection
- Gaming
- Social media

Amazon DynamoDB is built for speed, as it processes data at single-digit millisecond levels. There is also 99.999% availability when run across multiple regions.

Since it is serverless, Amazon DynamoDB is fairly easy to setup and manage. There is no need to handle version upgrades, cold starts or

patching.

Amazon CloudFront

Amazon CloudFront is a content delivery network (CDN) service. A CDN accelerates the delivery of static and dynamic web content. This is done by using sophisticated caching.

While there are various CDNs available, CloudFront has the advantage of being integrated with AWS services. There is also the benefit of thousands of relationships with telecom operators, as the network has hundreds of terabits of deployed capacity. There is also the integration with the network of AWS and its connected regions.

AWS Glue

AWS Glue is a serverless tool that prepares data for analytics, machine learning or AI projects. It has connections to more than 100 data sources.

With the tool, you can manage your data in a centralized hub. You can use visual tools to create, run and monitor the data pipelines. These can then be loaded into data lakes. AWS Glue also leverages generative AI to streamline the processes.

Amazon Redshift

Amazon Redshift is a data warehouse service. This is a centralized hub for handling large volumes of any kind of data. For Redshift, it's at petabyte scale. To put this into perspective, a petabyte can store more than 200,000 HD movies.

A major benefit of Redshift is cost-effectiveness. Compared to other cloud data warehouses, it can deliver up to 3X better price performance and 7x better throughput.

AWS CloudWatch

AWS CloudWatch is a sophisticated monitoring system for 70 AWS services. The metrics are based on the following:

Standard resolution

The data is updated every minute. This is set by default for prebuilt AWS metrics.

High resolution

This data is updated every second. This is for when you have a custom metric.

For all these metrics, AWS provides alarms and notifications.

Shared Responsibility Model

The shared responsibility model shows the division of responsibilities between AWS and the customer. This is for the security of the infrastructure and services.

Figure 2-3 shows a diagram of AWS's shared responsibility model.

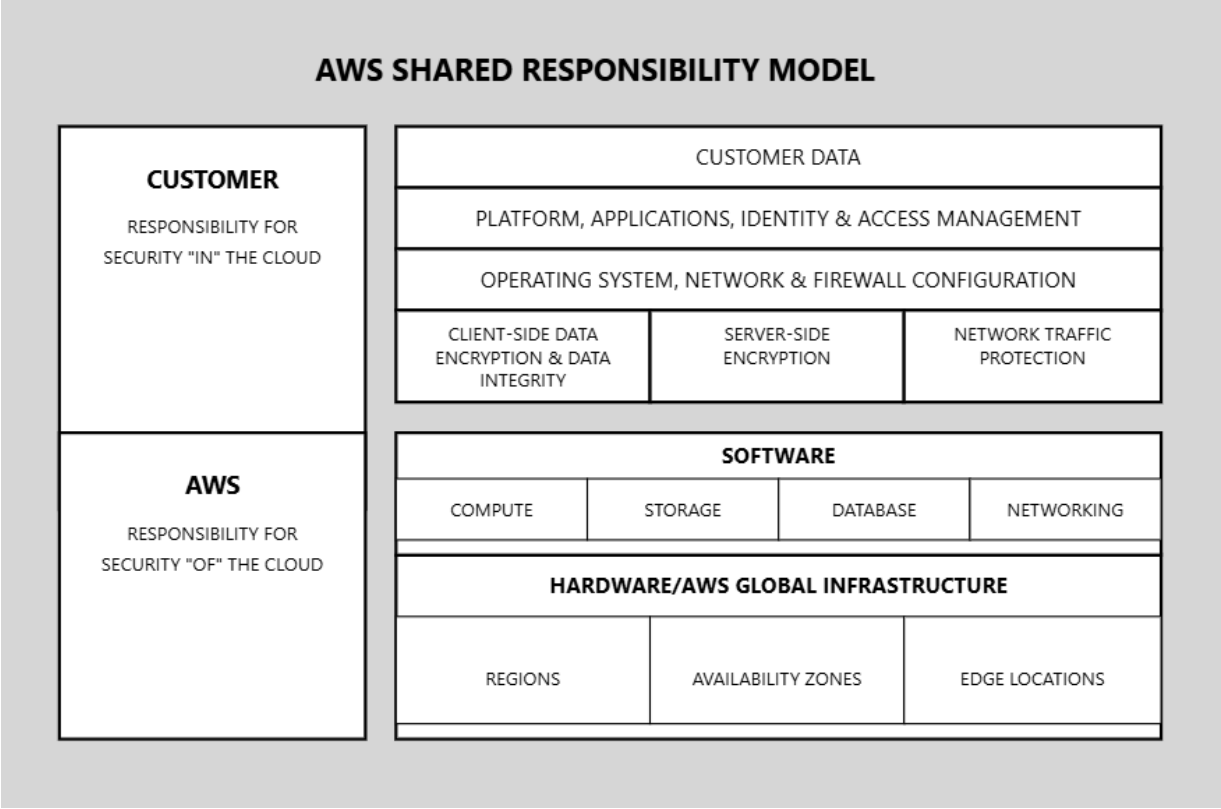


Figure 2-3. This is the AWS shared responsibility model.

For AWS, its responsibility is for "Security of the Cloud." It must protect the core infrastructure that runs the services. This is for physical servers,

storage and networking systems in the data centers. But this also includes the regions, AZs and Local Zones.

The customer’s responsibility, on the other hand, is about “Security in the Cloud.” This is based on numerous factors like AWS services used, the regions, laws and regulations, and the IT environment. AWS provides the security tools and the customers need to configure and manage them

For example, suppose you want to setup an EC2 instance. With this, the customer will take on the greatest responsibility. They will need to manage the OS updates, security patches, and configuration of firewall settings via security groups.

But with services like S3 or DynamoDB, there is less responsibility. The customer will manage tasks like access control, encryption policies, and permissions.

AWS Identity and Access Management (IAM)

AWS Identity and Access Management (IAM) manages the authentication and authorization of your AWS resources. There is much customization for the permissions. They can be broad, or strictly based on the principle of least privilege.

Here are the different types of IAM users and entities:

IAM Users

These are persons, apps or systems. Each has a name and credentials. If an IAM user has administrator privileges, this is not also a root user. Rather, this user has full administrative control over AWS services.

IAM Groups

This is a group of IAM users. An AIM group allows you to set permissions for multiple users. For example, a Developer Group may have permissions to use S3 storage. So when there is a new member added, they will have this permission. Or, if a person is removed from the group, the permissions will be terminated.

IAM Roles

This is not a user. It’s an identity that has specific permissions. For example, a role can be assigned to one or more IAM users. Typically, a role is for granting temporary access to AWS resources.

Another layer of security of IAM is Multi-Factor Authentication (MFA). This requires two or more proofs of identity to log in. An example would be a password and a code from your smartphone or an authentication app.

A key part of IAM is establishing policies. These are in JSON format and define who can do what. There are two main types of policies:

Identity-based policies: You will assign these to users, groups, or roles.

Resource-based policies: You will assign these to specific resources, like an S3 bucket.

AWS provides tools to customize and test your IAM setup. For example, IAM Access Analyzer helps you identify if any of your resources are shared with outsiders. In some cases, you may not even realize this is happening.

Setting Up AWS

Let’s go through the process of setting up AWS:

1. Go to aws.amazon.com and select “Create an AWS Account,” which you’ll find on the top right side of the screen.
2. Enter a user name, email address and password. You will then need to confirm your email.
3. You’ll add details like your full name, phone number, and address. You’ll also choose an account type, which is either Personal or Business.
4. Enter valid credit or debit card information for billing purposes. AWS requires this even if you plan to use the Free Tier.

5. Select a verification method – such as a text message or voice call – to receive a PIN. Enter the PIN to verify your number.
6. Choose a support plan that fits your needs. The Basic plan is free and suitable for most new users.

When you sign into the account, you will be taken to the AWS Management Console. You can see this in Figure 2-4.

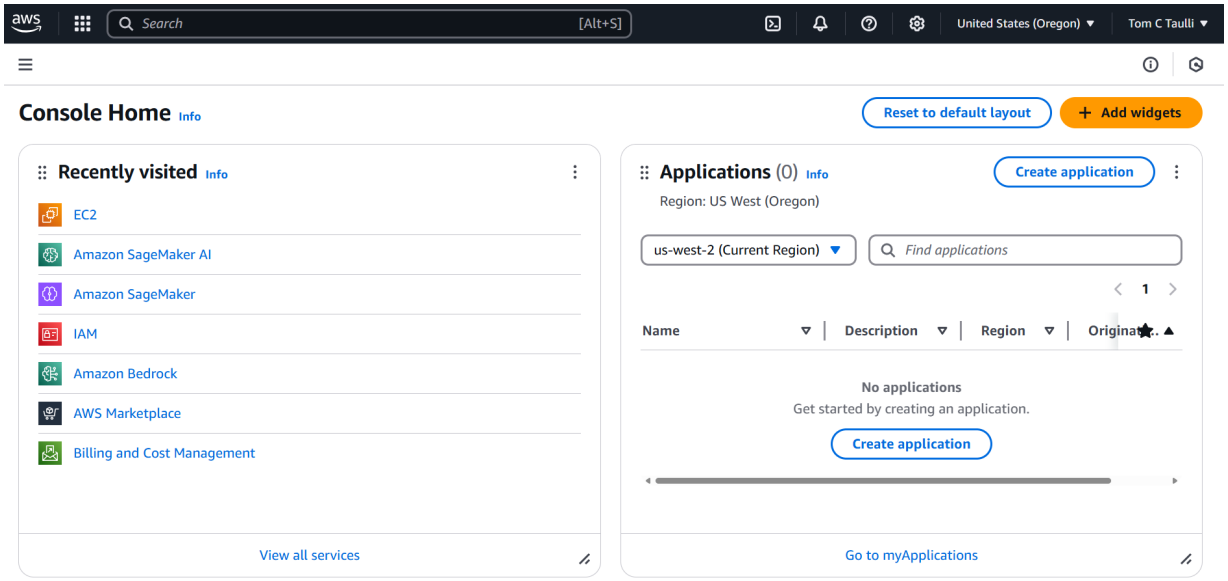


Figure 2-4. This is the AWS Management Console.

Here are some of the widgets on the AWS Management Console:

Available services

If you click the square icon at the top-left of the screen, you will see a dropdown that has a list of AWS services. There is also a search box for this.

Recently visited

These are AWS services you have used. This can be a convenient way to navigate AWS.

Applications

This shows your applications and the regions they are located. You can also create a new application here.

Welcome to AWS

You will find helpful documentation, training materials and AWS updates.

AWS Health

This monitors issues with your resources.

Cost and usage

You will see the current cost for the month. There is also a forecast feature.

Security

This provides any alerts or suggestions about vulnerabilities.

For this dashboard, you can customize it by moving the widgets.

Conclusion

In this chapter, we covered the fundamentals of the cloud and the AWS platform. We learned about concepts like cloud models and cloud service types. We then looked at how AWS is organized and the various services. After this, we got a quick demo of AWS.

In the next chapter, we'll look at the foundations of AI and Machine Learning (ML).

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